

# EG1160 Chip User Manual

High voltage, high current half-bridge driver chip

Version Change Log

| Version number, | date, description                       |  |
|-----------------|-----------------------------------------|--|
| V1.0            | EG1160 Datasheet (Draft) - May 16, 2018 |  |
|                 |                                         |  |
|                 |                                         |  |

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# EG1160 Chip Datasheet V1.0

## 1. Characteristics

• High-end floating bootstrap power supply design, withstand voltage up to 600V  
 • Compatible with 5V and 3.3V input voltages  
 • Maximum frequency support 500kHz  
 • Low-side VCC voltage range 10V-20V  
 • Output current capability IO+/- 2A/2.5A  
 • High-side MOSFET cycle-by-cycle peak current protection  
 • Low-side MOSFET cycle-by-cycle peak current protection  
 • Built-in dead-time control circuit  
 • Built-in latch-up function, completely preventing simultaneous conduction of high-side and low-side MOSFET outputs  
 • HIN input channel high level active, controlling high-side HO output  
 • LIN input channel high level active, controlling low-side LO output  
 • Low-side power supply VCC undervoltage protection  
 • High-side power supply VB undervoltage protection  
 • Package: SOP16

## 2. Description

The EG1160 is a high-performance, cost-effective, multi-functional half-bridge driver chip that integrates a 5V linear power supply, operational amplifier, MOSFET peak current protection, VCC undervoltage protection, VB undervoltage protection, dead-time control circuit, latch-up circuit, level shifting circuit, pulse filter circuit, and output drive circuit. It is very suitable for use in power supplies, motors, and other applications.

The EG1160 operates at up to 600V on the high-side and has a wide power supply voltage range of 10V to 20V on the low-side (Vcc), with a static power consumption of less than 1.5mA. This chip features a latch-up function to prevent simultaneous conduction of the output power transistors. The input channels HIN and LIN have a built-in 200K pull-down resistor, which keeps the high- and low-power MOSFETs off when the input is floating. The output current capability is IO+/- 2A/2.5A, and it is packaged in an SOP16 package.

## 3. Application Areas

• Sine Wave Inverter  
 • Solar Controller  
 • Industrial Control System

• Digital Generator  
 • Brushless Motor Driver  
 • High Voltage Class-D Amplifier

4. Pins

4.1 Pin Definitions

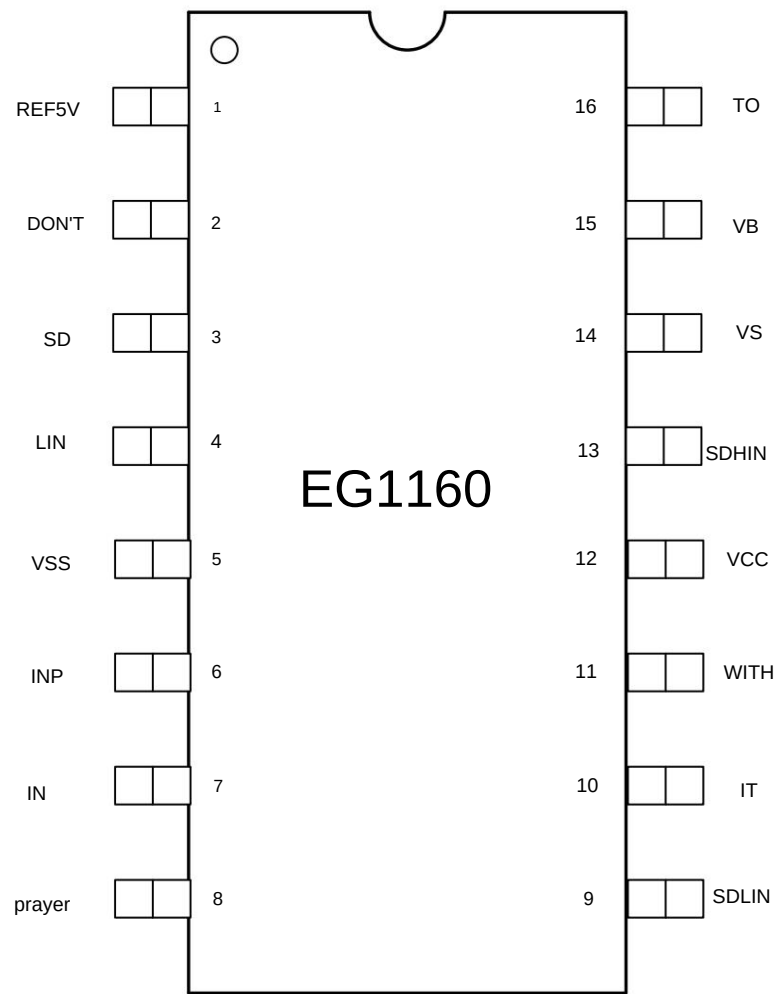


Figure 4-1. EG1160 Pin Definitions

4.2 Pin Description

| Pin number | Pin name I/O | describe                                                                                                                                                                                                          |
|------------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1          | REF5V        | 0 5V reference output, drive current 20mA.                                                                                                                                                                        |
| 2          | DON'T        | The logic input control signal is active high, controlling the on/off state of the high-side power MOSFET.<br>end<br>"1" indicates that the power MOSFET is enabled.<br>"0" means the power MOSFET is turned off. |
| 3          | SD           | A high level disables PWM output, while a low level enables PWM output.                                                                                                                                           |
| 4          | LIN          | The logic input control signal is active high, controlling the on/off state of the low-side power MOSFET.<br>end<br>"1" indicates that the power MOSFET is enabled.<br>"0" means the power MOSFET is turned off.  |
| 5          | VSS          | AGND is the chip's signal ground.                                                                                                                                                                                 |
| 6          | INP          | I. Op-amp input positive terminal                                                                                                                                                                                 |
| 7          | IN           | I. Op-amp input negative terminal                                                                                                                                                                                 |
| 8          | prayer       | O Op-amp output terminal                                                                                                                                                                                          |
| 9          | SDLIN        | I. Low-side MOSFET current comparator input port.                                                                                                                                                                 |
| 10         | IT           | O output controls the on/off state of the low-side MOS power transistor.                                                                                                                                          |
| 11         | WITH         | PGND is the chip's power ground.                                                                                                                                                                                  |
| 12         | VCC          | Power chip power supply, voltage range 10V-20V.                                                                                                                                                                   |
| 13         | SDHIN        | I. High-side MOSFET current comparator input port.                                                                                                                                                                |
| 14         | VS           | O High-end suspended ground terminal.                                                                                                                                                                             |
| 15         | VB           | Power high-end floating power supply.                                                                                                                                                                             |
| 16         | TO           | O output controls the on/off state of the high-side MOSFET power transistor.                                                                                                                                      |

## 5. Structural Diagram

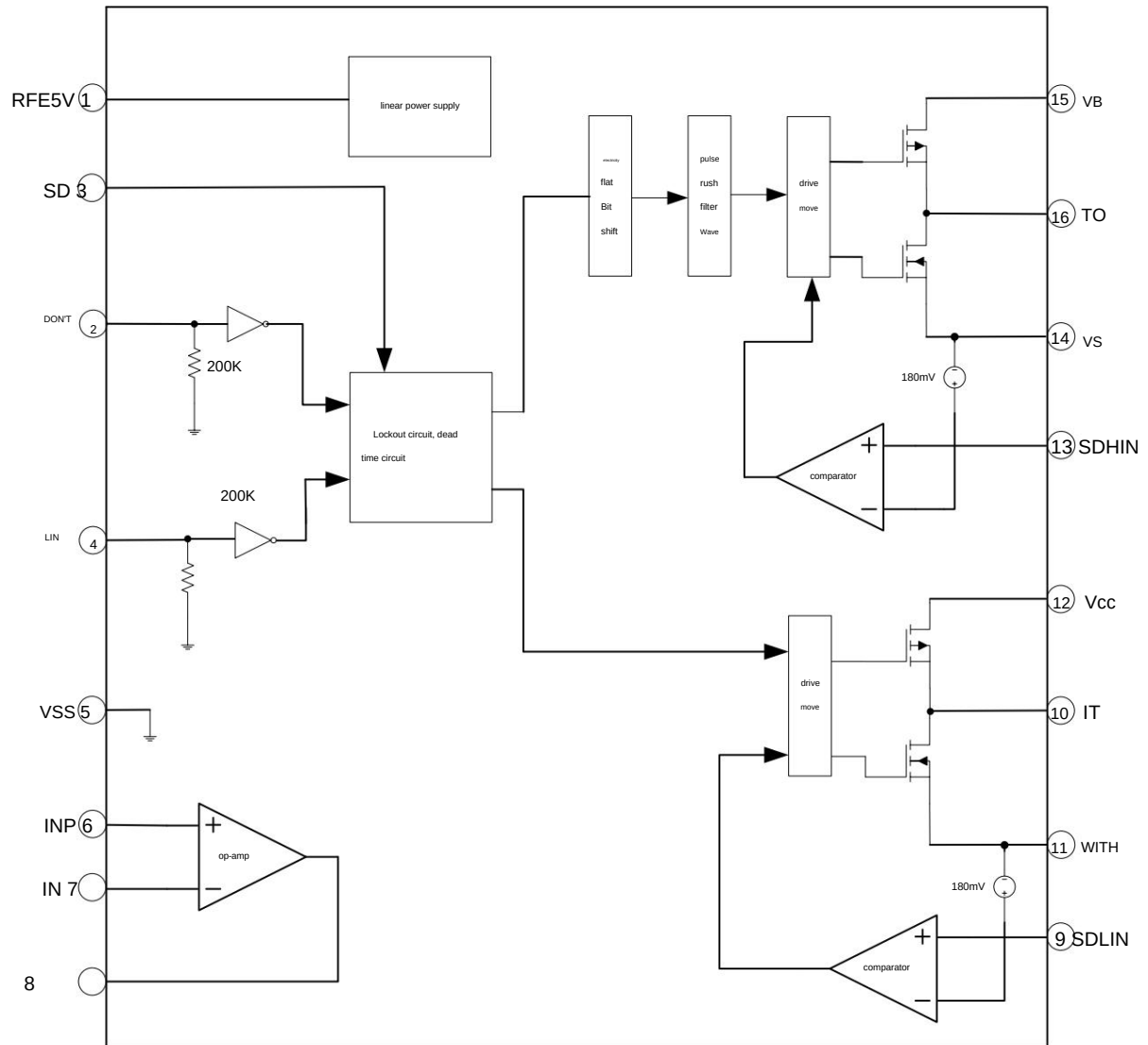


Figure 5-1. Internal circuit diagram of EG1160

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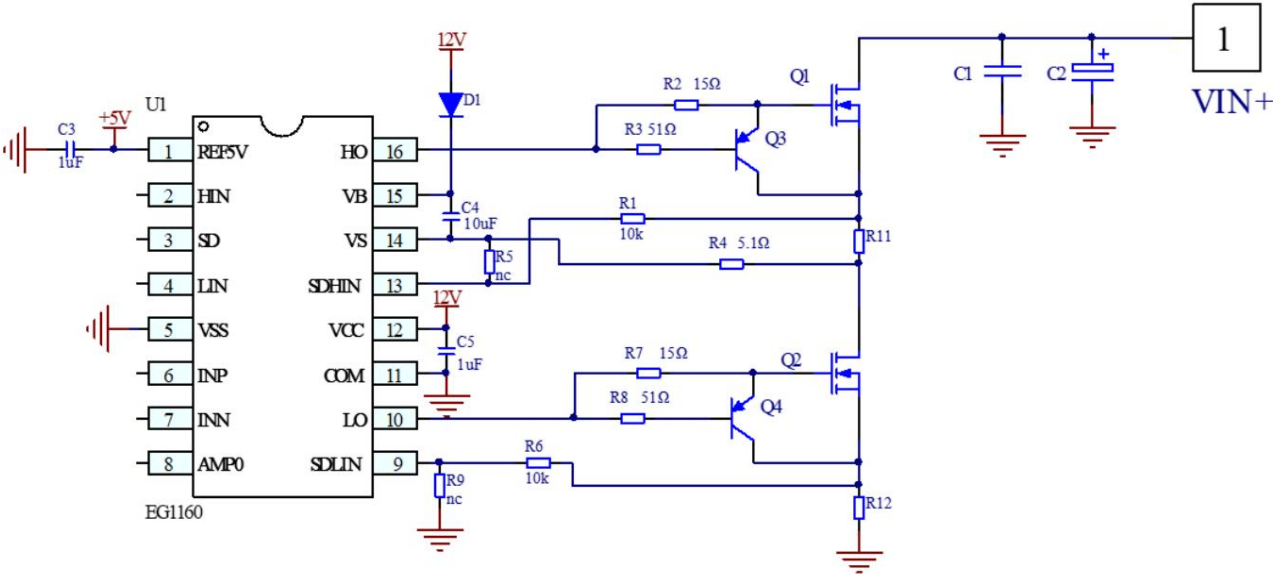


Figure 6-3. High-voltage, high-power applications

7. Electrical characteristics

7.1 Limiting Parameters

Unless otherwise specified, under the condition of TA=25y

| Parameter name symbol                   |           | Test conditions | Minimum and maximum units |         |
|-----------------------------------------|-----------|-----------------|---------------------------|---------|
| High-end output                         | TO        |                 | VS-0.3 VB+0.3             | In      |
| Bootable high-end VB power supply       | VB        |                 | -0.3                      | 600     |
| High-end suspended ground               | VS        |                 | VB-20                     | VB+0.3  |
| High-end comparator input               | SDHIN     |                 | VS-0.3                    | VS+5    |
| low-end power supply                    | VCC       |                 | -0.3                      | 20      |
| Low-end output                          | IT        |                 | -0.3                      | VCC+0.3 |
| Low-side comparator input               | SDLIN     |                 | -0.3                      | +5      |
| Low-voltage op-amp ports INP, INN, AMPO |           |                 | -0.3                      | +5.5    |
| Control port                            | SDHIN LIN |                 | -0.3                      | 20      |
| Ambient temperature                     | T         |                 | -45                       | 125     |
| Storage temperature                     | Tstr      |                 | -65                       | 150     |
| Welding temperature                     | TL        | T=10S           |                           | 300     |

Note: Exceeding the listed limits may cause permanent damage to the chip's internal components. Prolonged operation under extreme conditions may affect the chip's reliability.



## 7.2 Typical Parameters

Unless otherwise specified, under the conditions of TA=25y, Vcc=15V, and load capacitance CL=10nF.

| Parameter Name                                      | Symbol | High Voltage Power Supply      | Test conditions                | minimum | typical | maximum | unit |
|-----------------------------------------------------|--------|--------------------------------|--------------------------------|---------|---------|---------|------|
| VB                                                  |        | VB                             |                                |         | 10      |         | 600  |
| VCC power supply                                    |        | VCC                            |                                |         | 10      |         | 20   |
| Ivc                                                 |        | VCC static electricity<br>flow | VCC=12V                        |         |         | 1       | 1.5  |
| ONE                                                 |        | VB Static Current              | VBS=12V                        |         | 50      | 150     | a    |
| Input signal high potential Vin(H)                  |        |                                | HIN and LIN input signals 2.5  |         |         |         | In   |
| Input signal low potential Vin(L)                   |        |                                | HIN and LIN input signals -0.3 |         | 0       | 1.0     | In   |
| Input logic signal high<br>Flat current             |        | Yes(H)                         | Vin=5V                         |         |         | 20      | a    |
| input logic signal low<br>Flat current              |        | Yes(L)                         | Vin=0V                         |         | -20     |         | a    |
| VCC Turn-on voltage VCC (ON)                        |        |                                | VCC relative to VSS            | 7.5     | 8.5     | 9.3     | In   |
| VCC (Off Voltage)                                   |        |                                | VCC relative to VSS            | 7.1     | 7.9     | 8.7     | In   |
| VB is the on-state voltage.                         |        |                                | VB compared to VS              | 7.2     | 8.2     | 9.1     | In   |
| VB (Off Voltage)                                    |        |                                | VB compared to VS              | 6.7     | 7.5     | 8.3     | In   |
| Reference voltage                                   |        |                                |                                |         |         |         |      |
| 5V reference output                                 |        | REF5V                          | VCC is enabled, VCC=12V        | 4.6     | 4.8     | 5.0     | In   |
| Linear adjustment rate                              |        | ΔVREF                          | VCC=10V to 20V                 |         | 3       | 50      | mV   |
| Load regulation                                     |        | ΔVREF                          | IL=0 to 20mA                   |         | 5       | 50      | mV   |
| Maximum output current Iomax                        |        |                                |                                | 20      |         |         | m.a. |
| operational amplifier                               |        |                                |                                |         |         |         |      |
| Input offset voltage                                |        | IN THE VOICE                   |                                |         |         | 10      | mV   |
| Input voltage range VAICR                           |        |                                |                                | -0.3    |         | 5.3     | In   |
| Output voltage range                                |        | Vout                           |                                | 0       |         | REF5V   | In   |
| Output current capability                           |        | Iamp                           |                                | 10      |         |         | m.a. |
| Current comparator                                  |        |                                |                                |         |         |         |      |
| High-side current comparator SDHIN                  |        |                                | Relative VS voltage            | 160     | 180     | 200     | mV   |
| Low-side current comparator SDLIN                   |        |                                | Relative COM voltage 160       |         | 180     | 200     | mV   |
| Low-side output (LO) switching time characteristics |        |                                |                                |         |         |         |      |
| Enable delay                                        |        | Ton                            | See Figure 7-1                 |         | 280     | 400     | nS   |

|                                                   |           |                                           |     |     |     |    |
|---------------------------------------------------|-----------|-------------------------------------------|-----|-----|-----|----|
| Guan Yinshi                                       | Toff      | See Figure 7-1                            | -   | 150 | 200 | nS |
| Ascent Time                                       | Tr        | See Figure 7-1, CL=10nF                   | -   | 120 | 200 | nS |
| Descent time                                      | Tf        | See Figure 7-1, CL=10nF                   | -   | 80  | 100 | nS |
| High-end output HO switching time characteristics |           |                                           |     |     |     |    |
| Enable delay                                      | Ton       | See Figure 7-2                            | -   | 300 | 400 | nS |
| Guan Yinshi                                       | Toff      | See Figure 7-2                            | -   | 170 | 400 | nS |
| Ascent Time                                       | Tr        | See Figure 7-2, CL=10nF                   | -   | 120 | 200 | nS |
| Descent time                                      | Tf        | See Figure 7-2, CL=10nF                   | -   | 100 | 100 | nS |
| Dead time characteristics                         |           |                                           |     |     |     |    |
| DT                                                | Dead Time | See Figure 7-3.<br>No-load capacitor CL=0 | 100 | 160 | 220 | nS |
| Output MOS drive capability                       |           |                                           |     |     |     |    |
| LO, HO output current IO+                         |           | Vo=0V, Vin=VIH<br>PW $\geq$ 10 $\mu$ S    | 1.8 | 2   | -   | A  |
| LO, HO output sink current IO-                    |           | Vo=12V, Vin=VIL<br>PW $\geq$ 10 $\mu$ S   | 2   | 2.5 | -   | A  |

7.3 Switching time characteristics and dead time waveform diagram

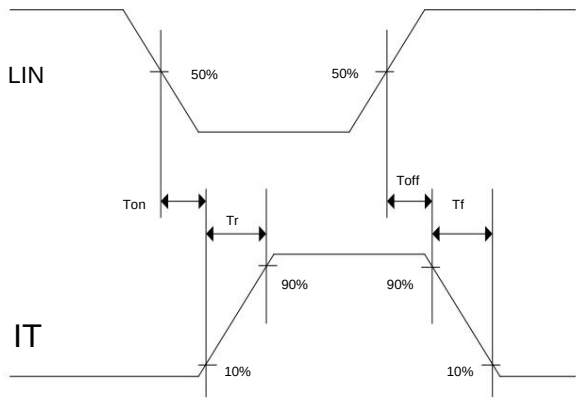
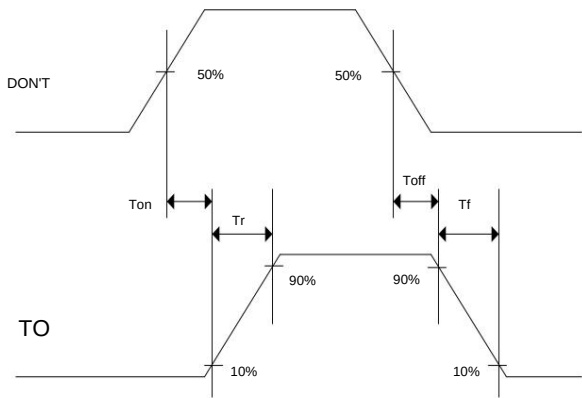


Figure 7-1. Waveform diagram of low-end output (LO) switching time.



7-2. High-side output HO switching time waveform diagram

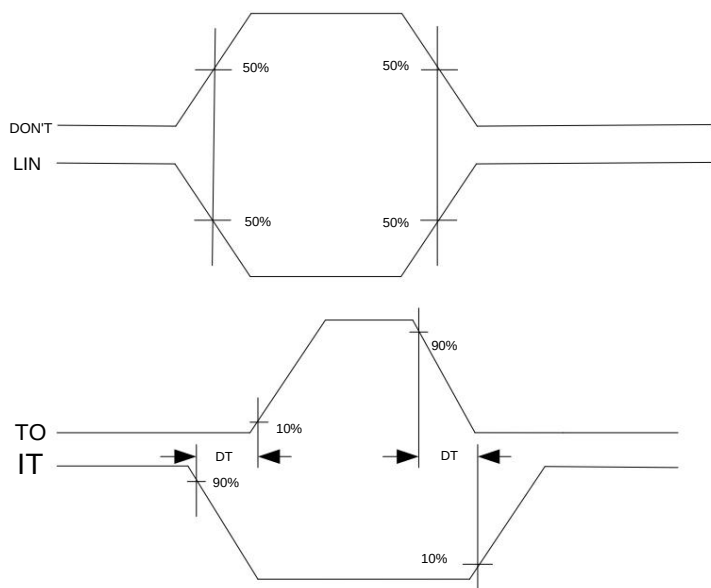


Figure 7-3. Dead time waveform diagram

## 8. Application Design

### 8.1 REF5V Input Capacitor

Placing a small-value high-frequency bypass capacitor between the REF5V pin and ground will reduce high-frequency noise at the REF5V pin. The high-frequency bypass capacitor can be selected from...

Place a 1uF ceramic capacitor as close as possible to the REF5V input pin of the chip during board layout.

### 8.2 VCC Low-end Power Supply

The EG1160 chip has undervoltage protection at the VCC low-side voltage to prevent insufficient VCC voltage from causing excessive turn-on resistance of the MOSFET, which could lead to...

The MOSFET burned out.

When laying out the PCB, make sure that the capacitors connected to ground (VSS, COM) of the VCC pin are as close to the chip as possible, and connect VSS and COM as close as possible.

### 8.3 VB High-End Power Supply

The EG1160 chip has undervoltage protection at the VB high-side voltage relative to VS to prevent excessive turn-on resistance of the MOSFET due to insufficient bootstrap capacitor voltage.

This can cause the MOSFET to burn out.

When laying out the PCB, make sure to place the bootstrap capacitors as close as possible to the chip pins.

### 8.4 Input Logic Signal Requirements and Output Driver Characteristics

The EG1160's main functions include logic signal input processing, dead time control, level conversion, floating bootstrap power supply structure, and upper/lower bridge diagrams.

Pole-type output. The logic signal input has a high-level threshold of 2.5V or higher and a low-level threshold of 1.0V or lower. The output voltage of the logic signal must be...

The low current allows the MCU output logic signals to be directly connected to the input channels of the EG1160.

The high-end upper bridge arm and low-end lower bridge arm output drivers have a maximum sink current of up to 2A and a maximum output current of up to 2.5A. The high-end upper bridge arm channel can...

It can withstand a voltage of 600V, with a small conduction delay between the input logic signal and the output control signal, and a low-side output turn-on conduction delay of 280ns.

The turn-off conduction delay is 150 ns, the high-side output turn-on conduction delay is 300 ns, and the turn-off conduction delay is 170 ns. The low-side output turn-on rise time...

The rise time for high-side output is 110ns, and the fall time for shutdown is 50ns.

The logic function diagram for the input and output signals is shown in Figure 8-4:

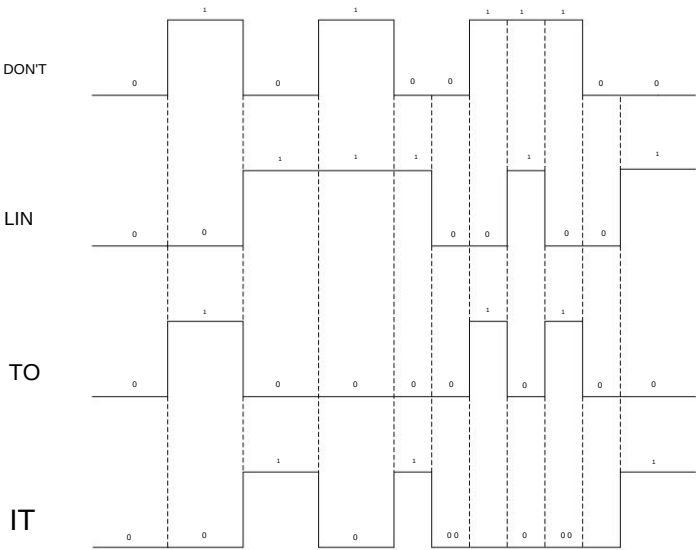


Figure 8-4. Logic function diagram of input and output signals

Input and output signal logic truth tables:

| enter                  |                                     | Output |   |
|------------------------|-------------------------------------|--------|---|
| Input and output logic |                                     |        |   |
| HIN (pin 2)            | LIN (pin 4) HO (pin 16) LO (pin 10) |        |   |
| 0                      | 0                                   | 0      | 0 |
| 0                      | 1                                   | 0      | 1 |
| 1                      | 0                                   | 1      | 0 |
| 1                      | 1                                   | 0      | 0 |

The truth table shows that when the input logic signals HIN are "1" and LIN are "0", the driver control output HO is "1", turning on the upper transistor.

When LO is "0", the lower transistor is turned off; when the input logic signals HIN are "0" and LIN are "1", the driver control output HO is "0", and the upper transistor is turned off.

When LO is "1", the lower transistor is turned on; when the input logic signals HIN and LIN are both "0" or both "1", the driver controls the output HO.

When LO is "0", both the upper and lower power transistors are turned off simultaneously; the internal logic processor prevents the upper and lower power transistors from being turned on simultaneously at the controller output, and has mutual blocking.

Lock function.

8.5 Bootstrap Circuit

The EG1160 employs a bootstrap floating drive power supply structure, greatly simplifying drive power supply design. It only requires a single power supply voltage, VCC, to drive both the high-side and low-side N-channel MOSFETs, significantly simplifying practical applications. The EG1160 can use...

An external bootstrap diode (as shown in Figure 8-5) and a bootstrap capacitor are used to automatically complete the bootstrap voltage boosting function. It is assumed that VC is activated during the period when the lower diode is on and the upper diode is off.

The bootstrap capacitor is charged to a sufficient voltage (VC=VCC). When HO outputs a high level and the upper transistor is turned on while the lower transistor is turned off, the voltage across the bootstrap capacitor is VC.

An equivalent voltage source is used as the power supply for the internal drivers VB and VS to drive the high-side N-channel MOSFET.

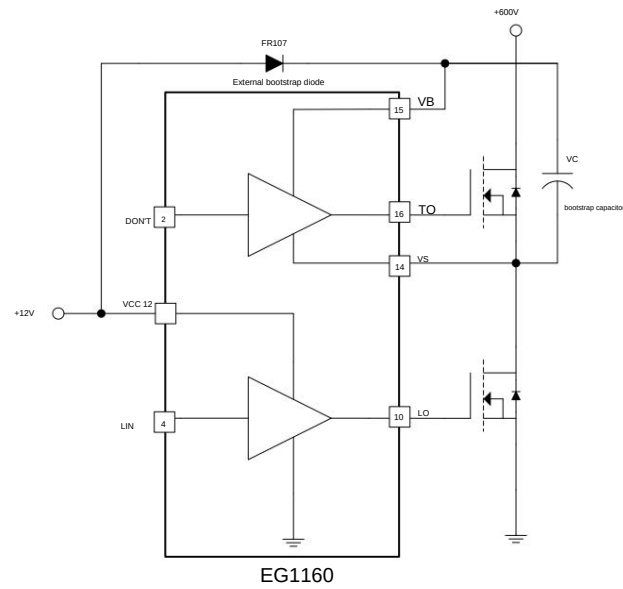
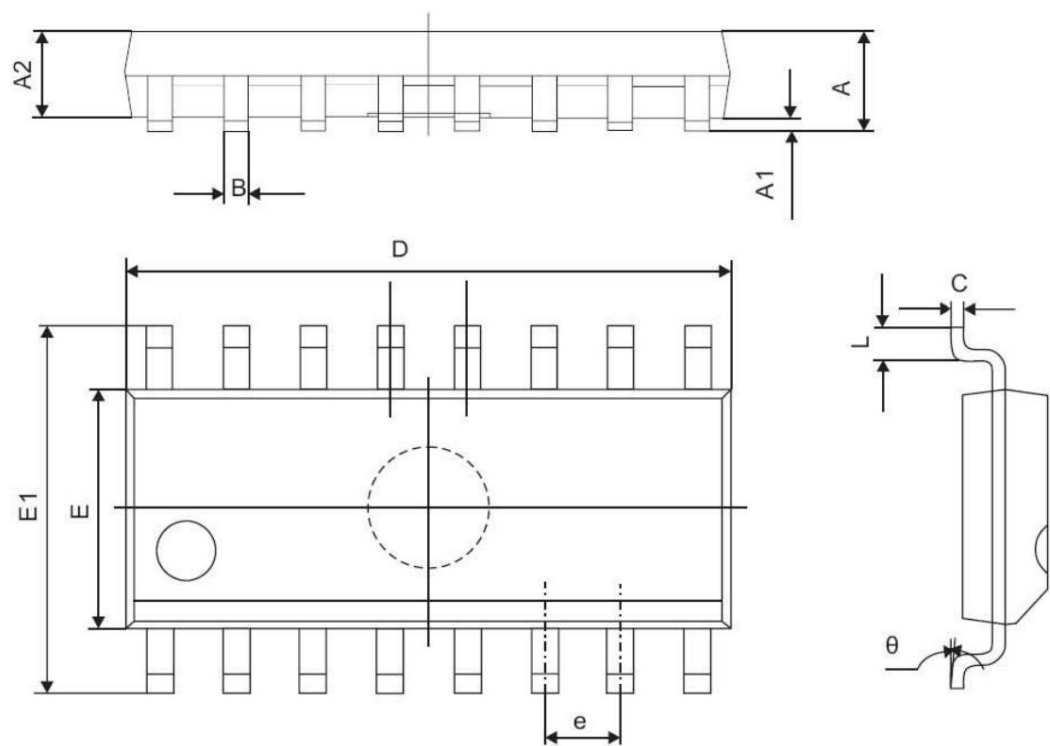


Figure 8-5. EG1160 bootstrap circuit structure

9. Package size

9.1 SOP16 package dimensions



| symbol | Dimensions (mm) |        |
|--------|-----------------|--------|
|        | Min             | Max    |
| A      | 1.350           | 1.750  |
| A1     | 0.100           | 0.250  |
| A2     | 1.350           | 1.550  |
| B      | 0.330           | 0.510  |
| C      | 0.190           | 0.250  |
| D      | 9.800           | 10.000 |
| and    | 3.800           | 4.000  |
| E1     | 5.800           | 6.300  |
| and    | 1.270 (TYPE)    |        |
| L      | 0.400           | 1.270  |
| I      | 0°              | 8°     |